

PREPARED FOR

Client: [Client Name]

Address: [Project Address]

Phone / Email: [Client Contact]

PROJECT

Replacement of two (2) existing V-configuration wood roof rafter sections with structural aluminum members — like-for-like appearance, equal or greater strength.

1. EXISTING CONDITIONS

The residence is an A-frame type structure with exposed sloped roof rafters ("V" members) supporting the roof deck at the gable overhangs. The members scheduled for replacement are solid wood timbers with an actual cross-section of 3½" × 5½" (nominal 4×6). Two (2) V-sections are to be removed and replaced. Each V-section consists of two sloped members approximately 20'-0" long meeting at the ridge, cantilevering approximately 1'-0" past the support beams to form the eave overhangs. The members carry wood plank decking with a metal roof panel above, both of which will be re-fastened to the new aluminum members.

2. SCOPE OF WORK

- 1. **Mobilization & protection:** Set up ladders/scaffolding as required; protect roofing, glazing, decking and landscaping in the work area.
- 2. **Temporary support:** Install temporary shoring to safely carry the roof load while each V-section is out.
- 3. **Demolition:** Carefully cut free and remove the two (2) existing wood V-rafter sections (four sloped members total, approx. 20'-0" each). Haul away and legally dispose of debris.
- 4. **Materials:** Furnish new **6061-T6 structural aluminum rectangular tube** supplied by **Eastern Metal Supply** (size per Section 3; each 20' member from a single full-length extrusion — no mid-span splices).
- 5. **Fabrication:** Miter-cut members at the field-verified ridge angle. Fabricate two (2) concealed internal chevron-plate inserts: 3/8" 6061-T6 plates waterjet/plasma-cut in one piece to the exact ridge angle, joined with welded spacers, pre-drilled and tapped — each insert slides into both tubes as a single unit (see DWG-3). Welds per AWS D1.2.
- 6. **Finish:** Finish aluminum in [Dark Bronze powder coat / wood-grain finish / paint to match] to visually match the existing dark-stained timbers.
- 7. **Installation:** Set new aluminum members in the exact position of the removed timbers. Ridge joint assembled with 1/4"-20 stainless flat-head countersunk screws, flush and concealed. **Members fully sealed: welded/snap-in aluminum end caps at the eave tips, sealed watertight**, with 1/4" weep holes at low points. Isolate aluminum from treated wood and dissimilar metals with barrier tape/washers to prevent galvanic corrosion. Seal miter joint with wood-tone sealant.
- 8. **Roof deck & metal panel:** Re-install wood plank decking over the new members with #14 winged self-drilling wood-to-metal screws (410 SS or ceramic-coated, concealed under roofing), and metal roof panel with #10 woodgrip screws with EPDM washers; drip edge at eaves and ridge cap at the peak (see DWG-4).
- 9. **Closeout:** Remove shoring, touch-up finish, final cleanup and walkthrough with owner.

3. MATERIAL SELECTION — STRENGTH MATCH

There is no aluminum tube produced in the exact $3\frac{1}{2}" \times 5\frac{1}{2}"$ timber dimension, so the closest standard extrusions stocked by Eastern Metal Supply were evaluated. The table compares each candidate against the existing wood member (Southern Pine No. 2, $F_b = 1,000$ psi, $E = 1.4 \times 10^6$ psi; allowable aluminum bending stress 19.5 ksi per Aluminum Design Manual, $E = 10.1 \times 10^6$ psi):

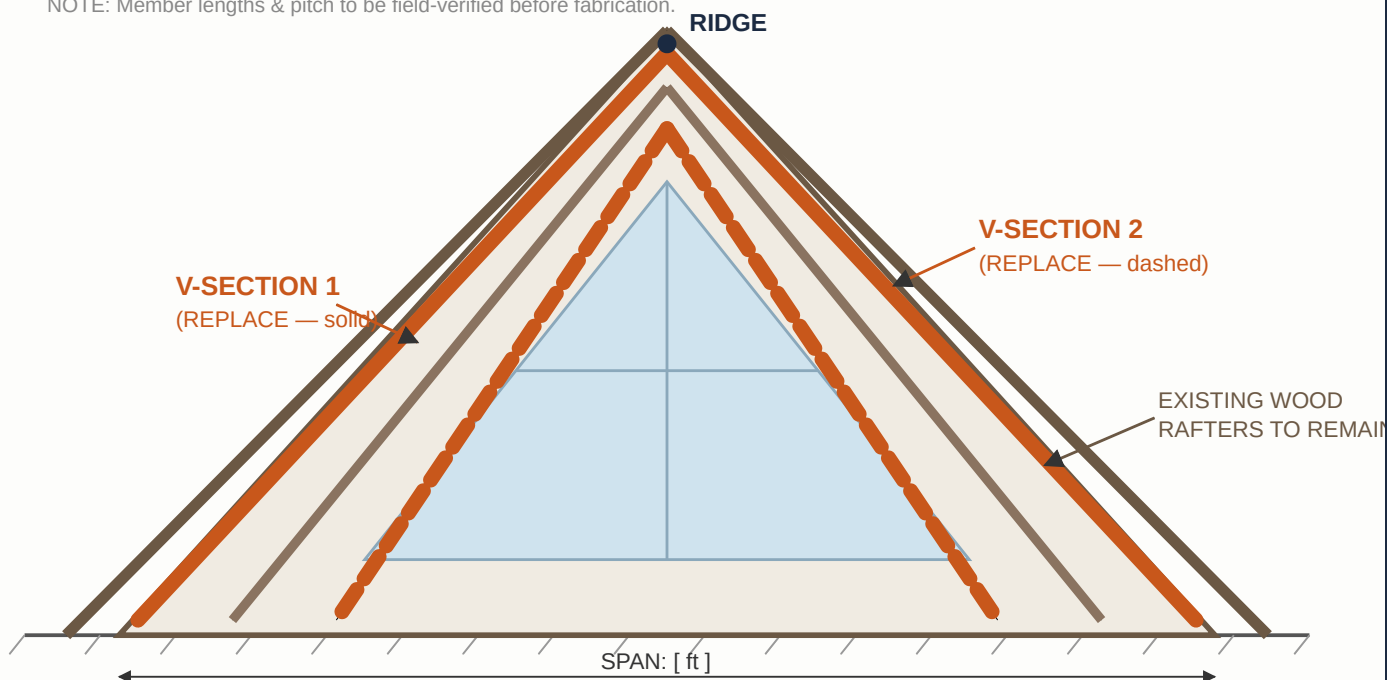
Member	Weight (lb/ft)	Stiffness EI (lb-in ²)	Stiffness vs. wood	Bending capacity vs. wood
Existing wood 4×6 (3.5"×5.5")	4.8	6.79×10^7	1.00×	1.00×
Aluminum 3"×5"×1/8" 6061-T6	2.3	6.76×10^7	0.99×	3.0×
Aluminum 3"×5"×3/16" 6061-T6 SELECTED	3.4	9.71×10^7	1.43×	4.2×
Aluminum 4"×6"×3/16" 6061-T6 (heavier alternate)	4.2	1.84×10^8	2.7×	6.7×

Selected member: 3" × 5" × 3/16" wall rectangular tube, alloy 6061-T6 (Eastern Metal Supply). It keeps a slim profile close to the existing timber ($\frac{1}{2}"$ smaller in each direction), provides **1.4× the stiffness and over 4× the bending strength** of the wood member it replaces, welds cleanly at 3/16" wall, and weighs 30% less than the timber. The 3/16" wall was chosen over 1/8" because it keeps the assembly stiffer than the original member with comfortable margin even if the existing lumber were a stiffer species (e.g. Douglas Fir).

4. DRAWINGS

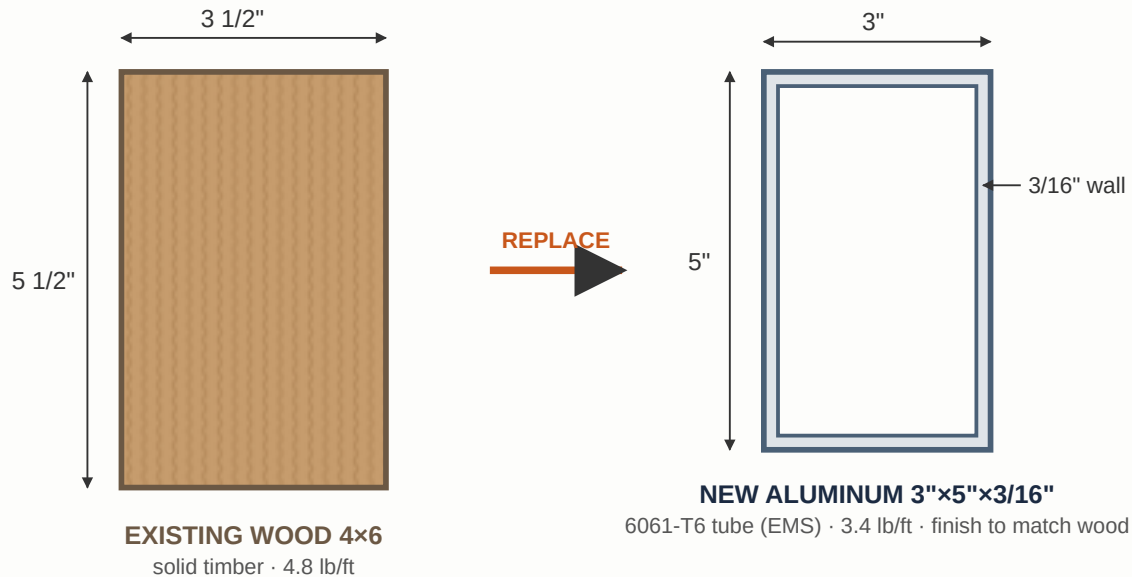
DWG-1 · GABLE ELEVATION — LOCATION OF V-SECTIONS TO BE REPLACED (N.T.S.)

NOTE: Member lengths & pitch to be field-verified before fabrication.



Each "V-section" = one pair of sloped rafters meeting at the ridge. Exact members to be marked on site with owner prior to demolition.

DWG-2 · CROSS-SECTION — EXISTING WOOD vs. NEW ALUMINUM (SCALE ~1:4)

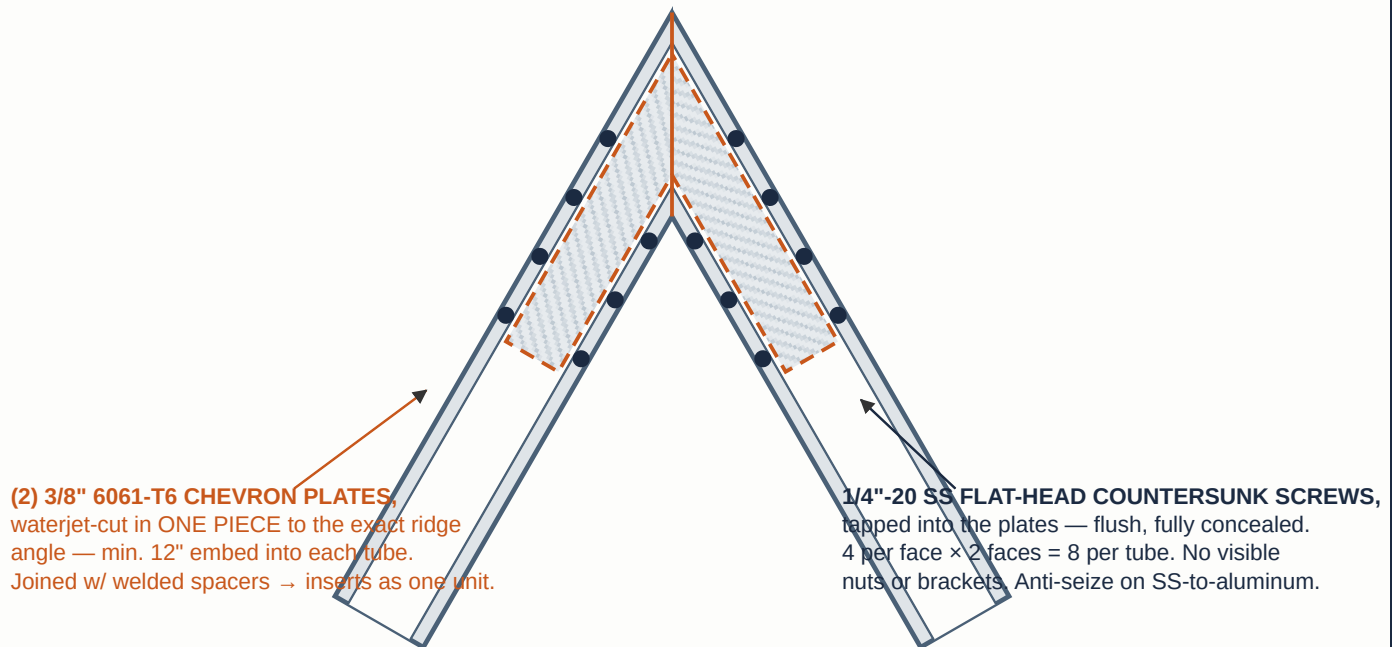


Strength: 4.2× bending capacity, 1.4× stiffness of existing timber — 30% lighter.

Heavier alternate available: 4"x6"x3/16" 6061-T6 (6.7× wood bending strength, 2.7× stiffness) if a bulkier profile is preferred.

DWG-3 · RIDGE JOINT — MITER-CUT TUBES W/ CONCEALED ONE-PIECE CHEVRON PLATES (N.T.S.)

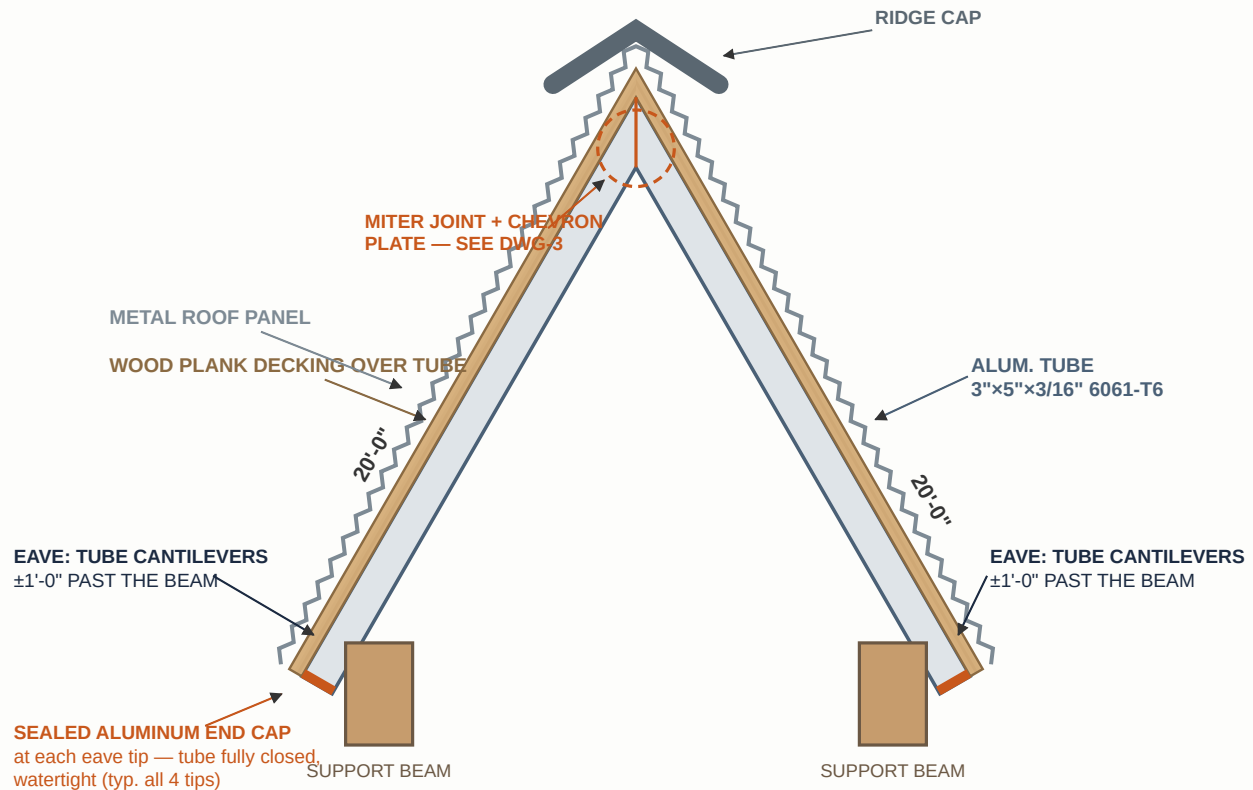
TUBES MITER-CUT & BUTTED TIGHT — HAIRLINE JOINT, WOOD-TONE SEALANT



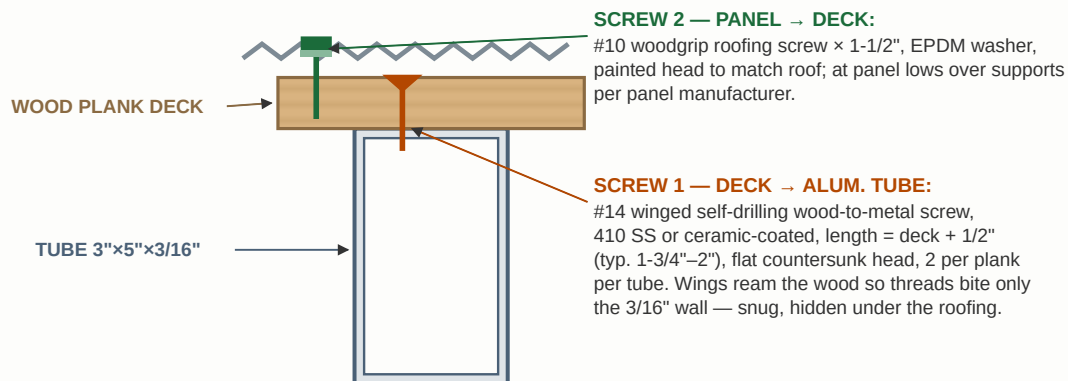
Joint capacity ≈ 8,000 lb per member (8 screws @ ~1,000 lb) — far above loads previously carried by the timber. 1/4" weep hole at each tube low point.

Bolted concealed joint keeps the 6061-T6 tubes at full strength (no weld heat-affected zone at the ridge). Field angle verified before plates are cut.

DWG-4 · GABLE FRONT VIEW — ROOF BUILD-UP, EAVE END CAPS & FASTENING (N.T.S.)



DETAIL B · DECK & PANEL FASTENING TO TUBE (section)



FASTENING NOTES:

1. No common zinc-plated screws against aluminum (galvanic corrosion) — 410 SS or ceramic-coated only; anti-seize on stainless.
2. Deck screws install from above and are concealed by the roofing panel — nothing visible from below.
3. Panel fastens to the wood deck (not directly to the tube) so roofing can be replaced later without touching the structure.
4. Drip edge at eave; sealed aluminum end caps at all four tube tips; 1/4" weep holes at tube low points.
5. Isolation barrier (butyl/bituminous tape) between pressure-treated wood and aluminum.

Deck planks and metal roof panel re-installed over the new aluminum members; all fasteners selected for aluminum compatibility.

5. PRICING

Item	Description	Amount
1. Mobilization, protection, scaffold rental & temporary shoring	Scaffold/lift rental, floor & glazing protection, shoring	\$ []
2. Demolition & disposal	Remove (2) wood V-sections, haul & dispose	\$ []
3. Aluminum material — Eastern Metal Supply	(4) RT 3"x5"x3/16" 6061-T6 × 21', 3/8" chevron plates, end caps, SS hardware	\$ []
4. Finish	Powder coat / finish to match existing timber color	\$ []
5. Fabrication	Miter cuts, chevron-plate inserts, drill & tap, weld spacers (AWS D1.2)	\$ []
6. Installation	Set members, concealed ridge joints, sealed end caps, isolation, sealants	\$ []
7. Roof deck & metal panel	Re-install wood deck (#14 winged screws) & roof panel (#10 EPDM), drip edge, ridge cap, cleanup	\$ []
Subtotal		\$ []
Sales tax ([%])		\$ []
TOTAL		\$ []

6. TERMS & CONDITIONS

- **Payment:** [50%] deposit to schedule and order material; balance due upon completion and owner walkthrough.
- **Schedule:** Approx. 2 weeks material & finish lead time + **4–5 working days on site with a 4-man crew** (one V-section demolished while the next is staged and installed), weather permitting.
- **Field verification:** All lengths, pitches and connection geometry will be field-measured before fabrication.
- **Warranty:** One (1) year on workmanship. Material per manufacturer/supplier warranty.
- **Exclusions:** Building permit & engineering letter (available as add-on if required by the county), repair of concealed rot/damage in members not listed, roofing repairs beyond the work area, painting of existing wood, electrical work.
- Any additional work will be quoted and approved in writing before proceeding (change order).

JRG Welding Corp — Authorized Signature / Date

Client Acceptance — Signature / Date